Caitlin Coffey

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Education

Bachelor of Science in Electrical and Computer Engineering, May 2023, GPA: 3.97/4.00
Franklin W. Olin College of Engineering, Needham, MA

- Recipient of a 4-year, 50% Olin Merit Scholarship
- **Relevant coursework:** Software Systems, Computer Architecture, Data Structures and Algorithms, Machine Learning, Principles of Integrated Engineering
- **Clubs:** Public Interest Technology, Olin Conductorless Orchestra, Society of Women Engineers

Work Experience

The Johns Hopkins University Applied Physics Laboratory, Laurel, MD
Cyber Security Intern 06/2022-Current

- Utilize Ghidra, Wireshark, Frida, GDB, and QEMU to find vulnerabilities in firmware and develop exploits.
- Build and develop automated data analysis scripts in Python, Bash, and C for reverse engineering methods.
- Research and create image processing models in TensorFlow & OpenCV.
- Work with and implement RF circuit designs for embedded systems.

The MITRE Corporation, Bedford, MA
Data Science & Network Security Intern 05/2021-08/2021

- Design, analyze, and publish an automated data processing pipeline using Apache Airflow and Apache Spark.
- Research and build recurrent neural network models in Python using TensorFlow, Keras, and Scikit-learn.

Qualcomm Inc., Boxborough, MA
Software Engineering Intern 05/2020-08/2020

- Automate and parallelize GPU performance verification tests using Python and multiprocessing libraries.
- Create a software tool to analyze GPU performance in Python using Matplotlib, NumPy, and Pandas.

Olin College of Engineering, Needham, MA
Software Design & Modeling and Simulation Teaching Assistant 09/2020-12/2021

- Assist in teaching data analysis/data science concepts. Analyze and debug code in Python and MATLAB.

Volunteer Experience

Women without Borders, Vienna, Austria
Software Developer 1/2022-05/2022

- Build and debug HTML, CSS, and JavaScript code on a front-facing website. Navigate WordPress & edit custom-built themes. Develop social media campaign material for multilingual, cross-cultural work.

Relevant Skills

Python – Object-oriented programming, artificial intelligence (image processing, natural language processing).

C, C++, Assembly - Reverse engineering, object-oriented programming, embedded systems programming (on Arduino, Raspberry Pi), device emulation, firmware modification (in ARM, x86).

Linux – System navigation, shell scripting, container development with Docker, version-control management with Git.